Medhansh Garg

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Education

University of Illinois, Urbana-Champaign Aug. 2023 – May 2027 (Expected)
- B.S. in Computer Engineering | Coursework: Data Structures & Algorithms, Computer Systems Engineering, Computer Architecture, Operating Systems, Distributed Systems, Computer Networks, Databases

Skills

- **Security**: Burp Suite, Nmap, Wireshark, Frida, JADX, Hashcat, OAuth2/CILogon SSO, RBAC, Row-Level Security, TOTP, sandboxing
- **Languages**: Python, TypeScript/JavaScript, C, C++, Java, Kotlin, Go, SQL
- **Web & Backend**: React, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, REST APIs
- **Infra**: Linux, Docker, Kubernetes, Azure, GitHub Actions CI/CD, nginx, Tailscale, MQTT, systemd, Git

Experience

Care Health Insurance, Gurugram, India | *Cybersecurity Intern* Jun. 2025 – Aug. 2025
- Mapped the insurer's attack surface, discovering **60+ unauthenticated internet-exposed services**; **30** pulled offline after my report
- Reverse-engineered WAF request signing into a Burp Suite replay exploit, proving signed requests could be forged
- Bypassed SSL pinning and root detection in the customer Android app (Frida, JADX), enabling live traffic interception and analysis

CS 124 Honors, Urbana, IL | *Lead Web Developer* Aug. 2025 – Present
- Lead an **11-person** team on the course platform for **300+ students** and **30+ staff**; implemented CILogon SSO with **6 least-privilege roles** and role preview without privilege escalation
- Routed all database access through a server-side API, keeping credentials out of the browser; designed a **246-test** suite blocking any release that breaks access rules
- Engineered attendance check-in on one-time codes rotating every **30s**, restricting check-in to students physically present

Hopcharge, Gurugram, India | *Software Engineering Intern* Jun. 2026 – Aug. 2026
- Developed VOLT.CV, a hiring dashboard for real candidate data, with database-layer row-level security on Google identities
- Shipped the Ad Engine, an AI ad pipeline for Meta and YouTube; scaled output **8x** (2–3/week to **3/day**)
- Consolidated **5 manual scripts** into Grid Ops, a FastAPI/React demand-forecasting service deployed behind nginx on a cloud VM

IoT++, Remote | *Software Engineering Intern* Jun. 2024 – Aug. 2024
- Developed real-time fire/smoke detection, **62%** more accurate than base YOLOv8; deployed to **12 edge devices** in a factory pilot
- Trained Random Forest and RNN models routing trucks to weighbridges from live traffic and equipment throughput

Projects

Mainframe: Face Unlock & Homelab | *Raspberry Pi, PAM* hackoverflow404.github.io/Portfolio/experience?entry=mainframe
- Designed laptop face unlock with Ed25519-signed verdicts bound to host, one-time challenge and 30s window, preventing forged or replayed unlocks on a shared network
- Operate **19 services** on a Raspberry Pi since May 2025 (NAS, Home Assistant, camera recorder) with **zero open router ports**; access only via private Tailscale mesh

Pensieve: Sandboxed AI Agent | *Python, bubblewrap* hackoverflow404.github.io/Portfolio/experience?entry=pensieve
- Contained an AI note-filing agent: sandboxed on a copy with no access to originals, read-only mounts enforcing visibility, and a rule-based checker rejecting any run that alters original text (**76 tests**)
- Isolated each agent run into one revertible git commit limited to unedited files; violations reject the run and trigger a phone alert

TPS-L2: Spotify Wall Display | *JavaScript, Vercel, Redis, Raspberry Pi* github.com/HackOverflow404/TPS-L2
- Designed QR sign-in for keyboardless screens: HMAC-signed self-expiring device codes, single-use token handoff and rate limiting, so a photographed QR code cannot hijack a login

Clippy: Jailbroken Echo Show 5 | *C++, Kotlin, Android* github.com/HackOverflow404/EchoCameraStreamer
- Unlocked the bootloader via a MediaTek bootrom exploit; replaced Amazon's OS with a source-built LineageOS ROM; diagnosed and fixed a vendor-library ABI mismatch disabling the camera
- Converted its camera into a security camera with **18h** of rolling footage on a NAS

Certifications & Publications

NYUx MicroBachelors in Cybersecurity Fundamentals | *Certification* Jun. 2022 – Apr. 2023
- 9-course program, **A grade**: authentication and access control, network security, penetration testing

Evaluation of Leet Speak on Password Strength and Security | *Published paper, IJSRST* researchgate.net
- Cracked leaked passwords and their leet-speak variants with Hashcat; no significant crack-rate difference, so substitutions fool strength meters but not attackers

Biometric Authentication and Cybersecurity in the Digital Age | *Published review paper* medcrave.com
- Evaluated biometrics, passwords and tokens for online banking; layering biometrics with one-time passwords is strongest

Leadership & Involvement

Illinois MicroTech, Urbana, IL | *External Vice President* | *EOH Project Lead* | *HackerFab Automation Member* Oct. 2025 – Present
- Run external communications, overseeing the outreach team, marketing chairs, website and EOH chairs
- Led a **15-person** team building a 4-stage interactive MEMS exhibit for **20,000+** Engineering Open House visitors
- HackerFab: automating the wafer stage of a student-built chip-fabrication rig for autonomous full-wafer movement